



Global Country Data

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About

- Information on countries worldwide
- Compares and analyze trends

This comprehensive dataset provides a wealth of information about **all countries worldwide**, covering a wide range of indicators and attributes. It encompasses demographic statistics, economic indicators, environmental factors, healthcare metrics, education statistics, and much more. With every country represented, this dataset offers a complete global perspective on various aspects of nations, enabling in-depth analyses and cross-country comparisons.

Descriptions of variables:

country

- birth_rate: number of births per 1,000 people per year
- life_expectancy: average number of years a person is expected to live
- gdp: total economic output of the country

Motivation for Analyzing

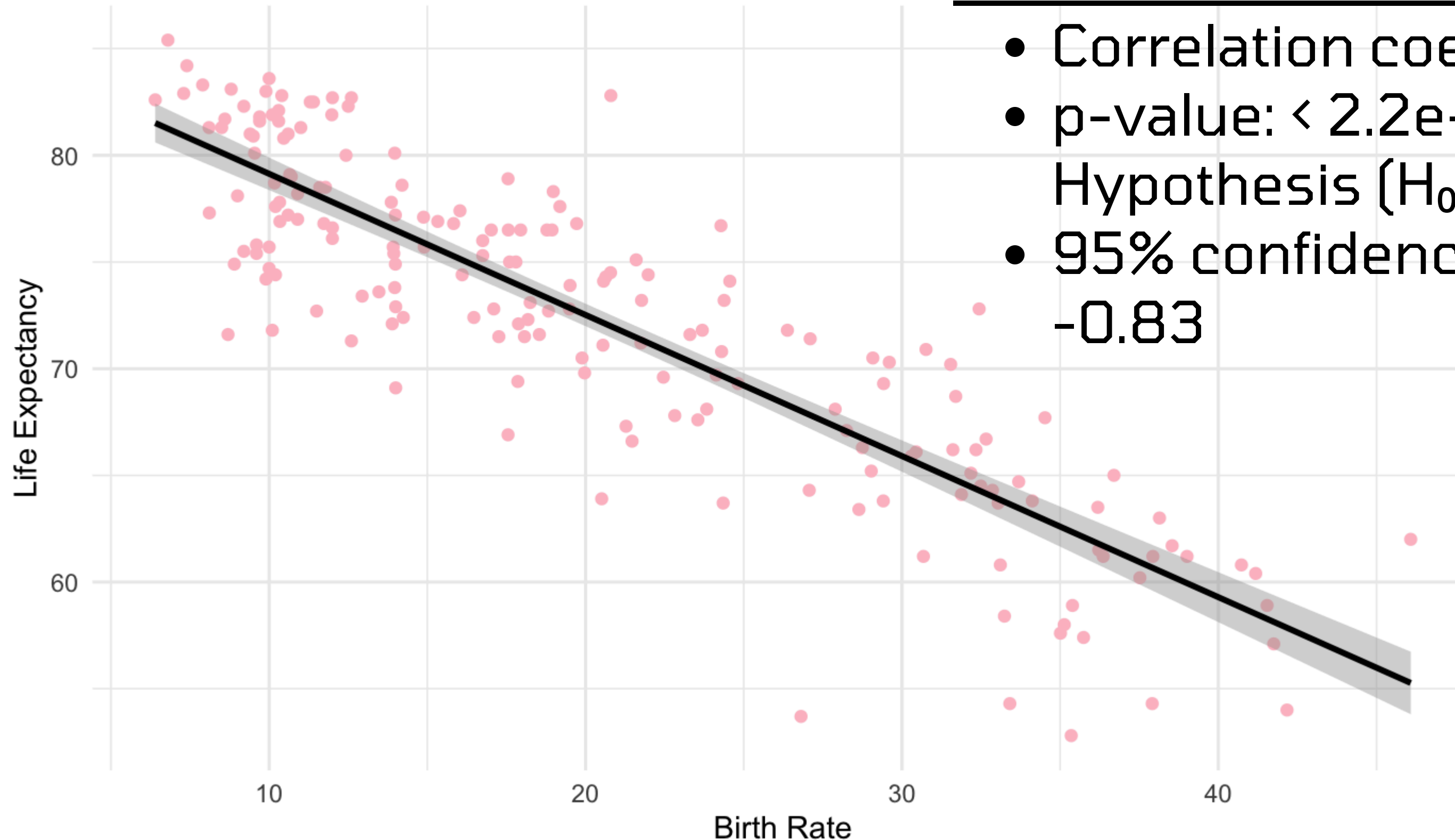
- Interesting

Analyzing the Relationship Between Birth Rate and Life Expectancy

Hypothesis: As birth rate increases, life expectancy decreases

Why? higher birth rates often coincide with challenges in healthcare, resources, and economic development

Life Expectancy vs Birth Rate

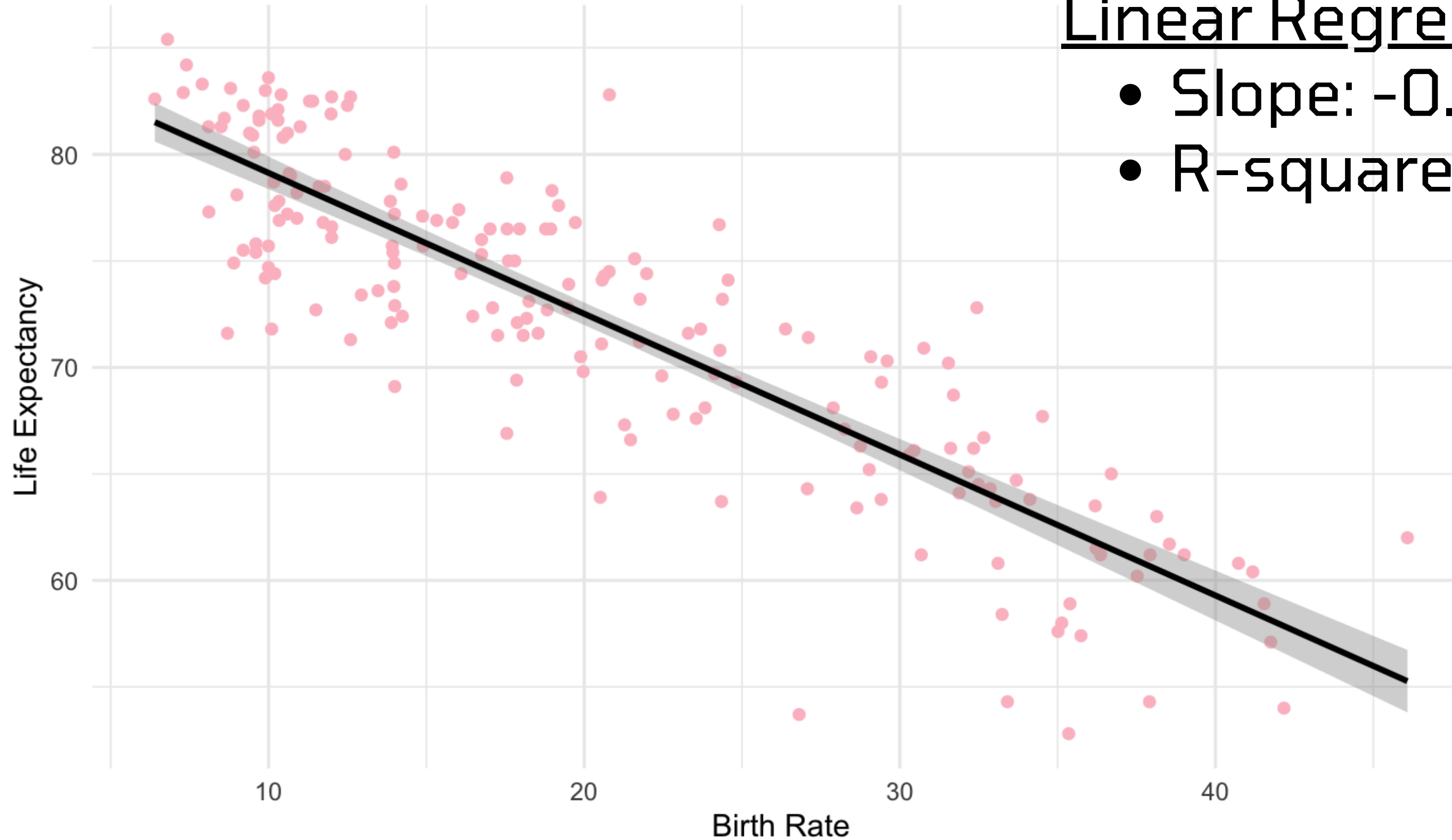


Pearson Correlation Test:

- Correlation coefficient: -0.87
- p-value: $< 2.2e-16$ (Reject Null Hypothesis (H_0))
- 95% confidence interval: -0.90 to -0.83

- There's a strong negative relationship between birth rate and life expectancy
- Relationship is statistically significant with a very low p-value

Life Expectancy vs Birth Rate



Linear Regression Model:

- Slope: -0.66
- R-squared: 0.7652

- Slope: For each unit increase in birth rate, life expectancy decreases by about 0.66 years.
- R-squared: 76.5% of the variation in life expectancy is explained by the birth rate.

Highest Birth Rate

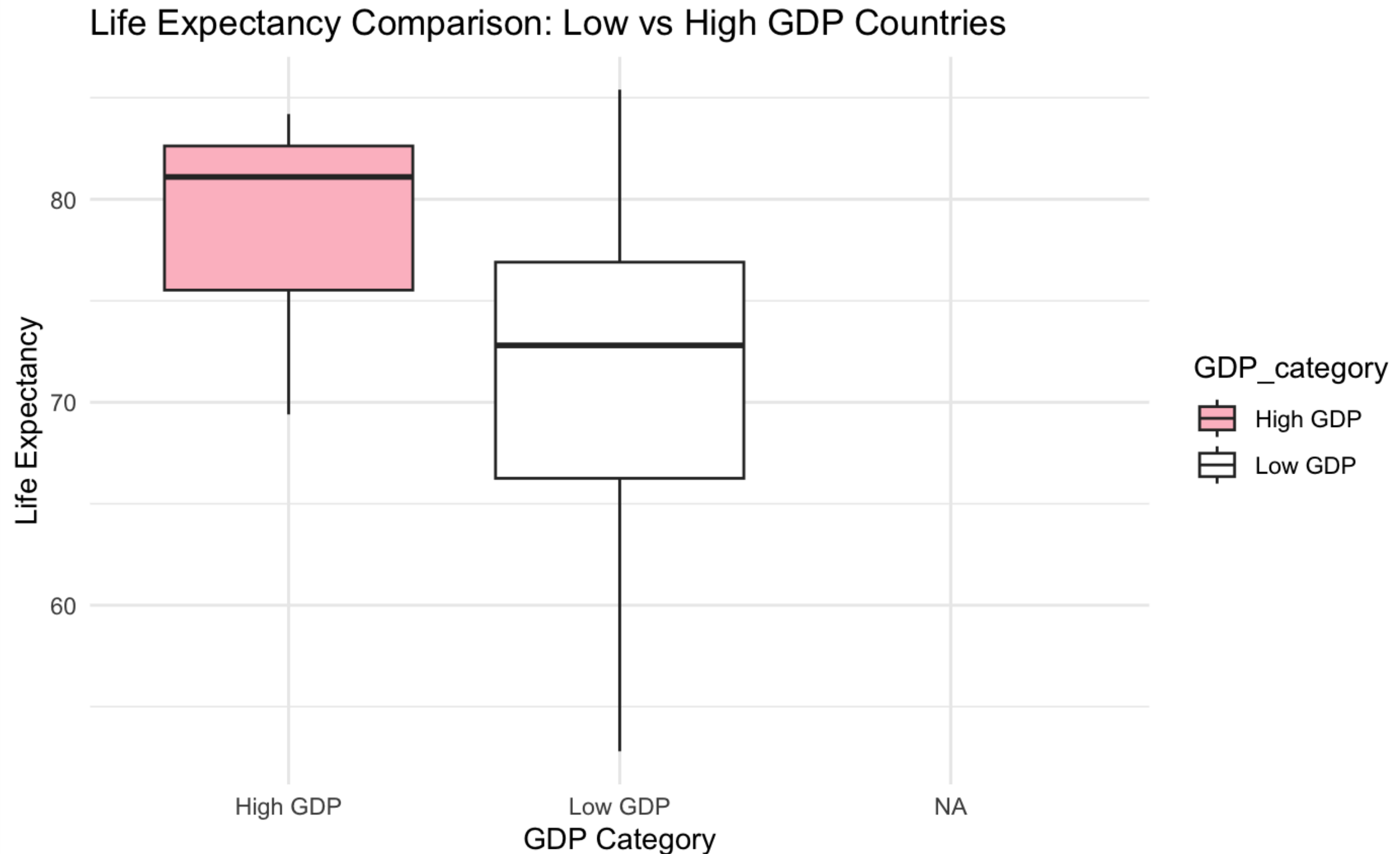
Country<chr>	Birth_Rate<dbl>	Life_expectancy<dbl>
Niger	46.08	62.0
Chad	42.17	54.0
Somalia	41.75	57.1
Mali	41.54	58.9
Democratic Republic of the Congo	41.18	60.4

Country<chr>	Birth_Rate<dbl>	Life_expectancy<dbl>
South Korea	6.4	82.6
San Marino	6.8	85.4
Italy	7.3	82.9
Japan	7.4	84.2
Spain	7.9	83.3

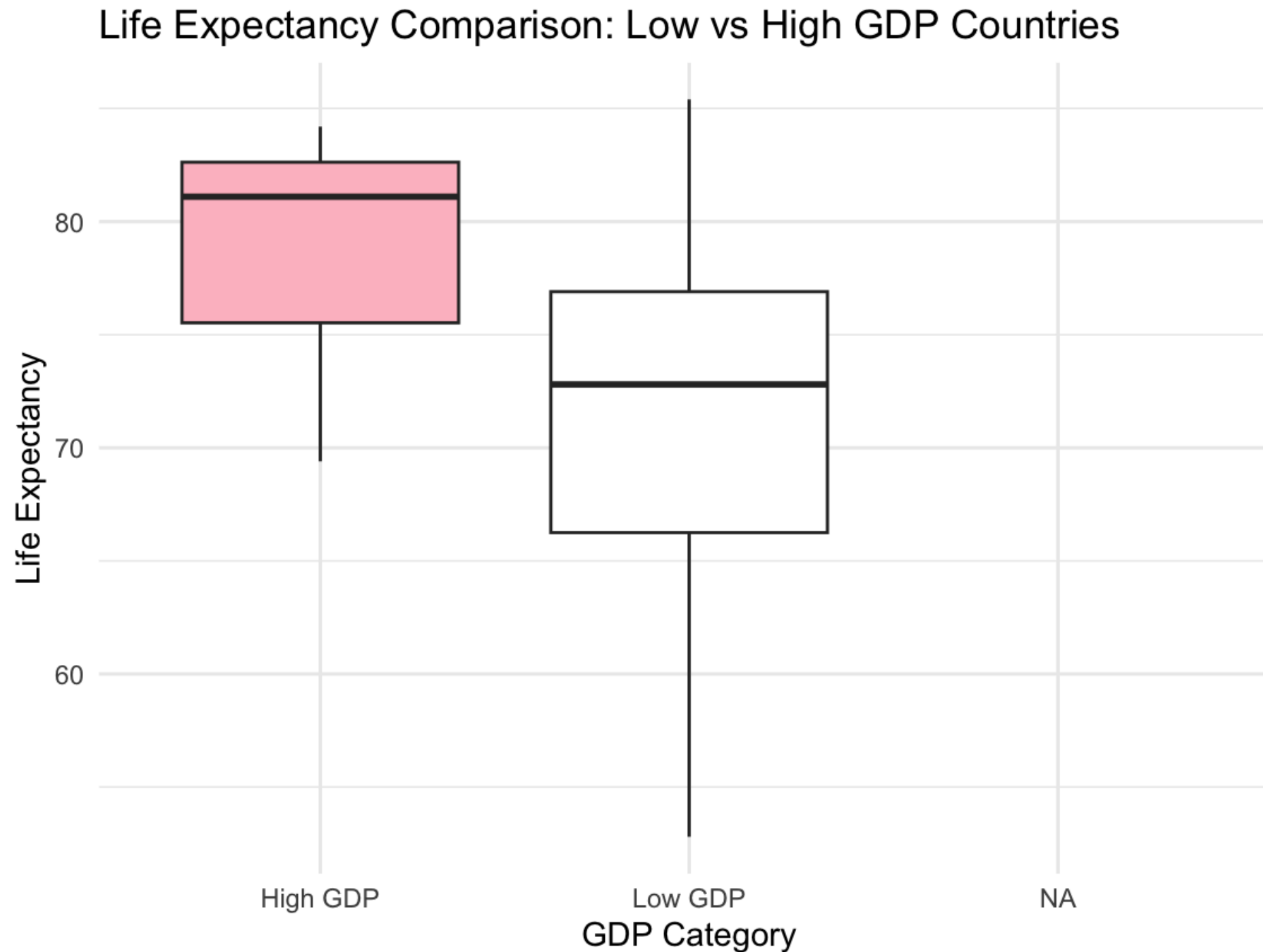
Lowest Birth Rate

Life Expectancy Comparison: High vs Low GDP

- p-value = 1.442e-05:
difference in life expectancy between high and low GDP countries is statistically significant
- reject the null hypothesis of no difference

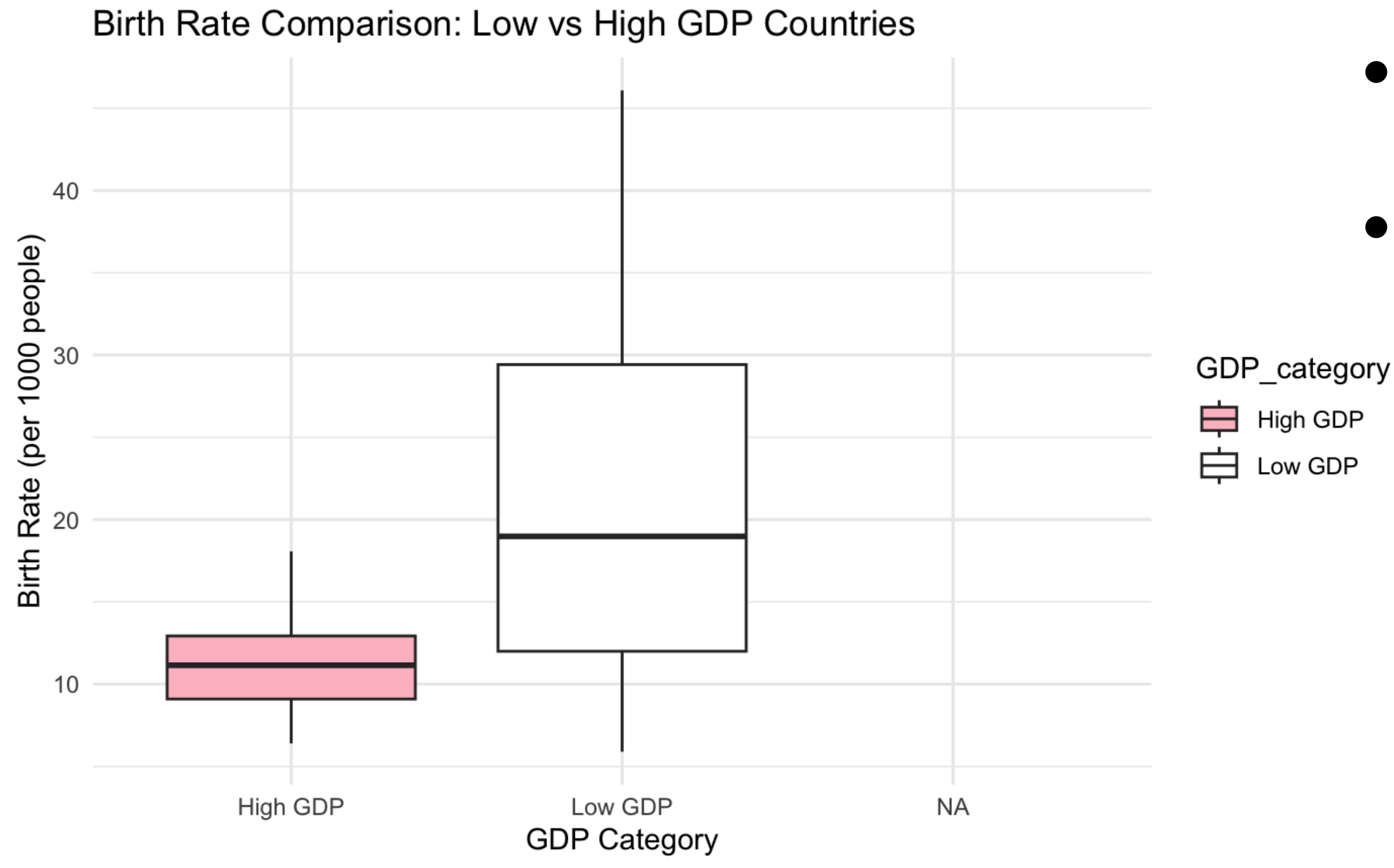


Life Expectancy Comparison



Mean life expectancy
in High GDP
countries: 78.88

Mean life expectancy
in Low GDP countries:
71.66



- $p\text{-value} = 1.1\text{e-}09$
- Mean in group High GDP: 11.5
- Mean in group Low GDP: 21

Country<chr>	Birth Rate<dbl>	GDP<dbl>
United States	11.6	2.142770e+13
China	10.9	1.991000e+13
Japan	7.4	5.081770e+12
Germany	9.5	3.845630e+12
United Kingdom	11.0	2.827113e+12

Highest GDP

Lowest GDP

Country<chr>	Birth Rate<dbl>	GDP<dbl>
Kiribati	27.89	194647202
Marshall Islands	29.03	221278000
Palau	14.00	283994900
Federated States of Micronesia	22.82	401932279

Conclusion

- Life expectancy is strongly linked to both GDP and birth rate.
- Countries with higher GDP → longer life expectancies & lower birth rates.
- Lower GDP → shorter life expectancies & higher birth rates
- This pattern indicates that wealthier countries typically offer better living conditions and healthcare, while also managing population growth more effectively, leading to longer life expectancies.
- However, it's important to remember that this is a correlation, not causation, and other factors may also play a role.